

AI CHATBOT RAG AI SYSTEM

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PROJECT 1

AI Chatbot

What It Does

A conversational AI chatbot for automated business support

- **Holds real conversations**

Remembers everything said earlier — users never need to repeat themselves

- **Always available**

Responds instantly, 24 hours a day, with no wait time or staffing cost

- **Fully configurable**

Businesses set the tone, focus, and behaviour through simple system prompts

- **Real-time responses**

Powered by Groq's fast AI engine for near-instant, accurate replies

The Business Problem It Solves

- **High support costs**

Hiring and training support staff is expensive — AI handles routine queries automatically

- **Slow response times**

Customers expect instant answers; manual systems create frustrating delays

- **Repetitive work**

Staff spend hours answering the same questions, taking time away from higher-value tasks

- **No after-hours support**

Human teams can't work 24/7 — AI is always on, for every time zone

Challenges & How We Solved Them

Challenge · API Integration

→ Studied the full request-response workflow and tested until the connection was reliable

Challenge · Conversation Memory

→ Built a structured message history that stores and replays the full chat context each time

Challenge · Response Quality

→ Tuned prompts and AI parameters until outputs were consistent and accurate

PROJECT 2

RAG Document AI

What It Does

- **Reads any PDF**

Upload reports, manuals, or policies — the AI reads and indexes them automatically

- **Finds answers by meaning**

Semantic search understands intent, not just keywords, for far more accurate results

- **Shows its sources**

Every answer includes the exact document section it came from — fully traceable

- **Remembers your session**

Maintains chat history so you can ask follow-up questions naturally

The Business Problem It Solves

- **Document overload**

Organisations store thousands of PDFs — finding one answer takes far too long

- **Inefficient search**

Staff scan files manually, wasting hours that could go to productive work

- **Knowledge locked away**

Critical information sits in documents that nobody reads or can easily find

- **Lost productivity**

Employees spend up to 20% of their time just searching for information

Challenges & How We Solved Them

Challenge · Retrieval Inaccuracies

→ Improved document chunking strategy and added step-by-step retrieval debugging

Challenge · Library Conflicts

→ Updated all dependencies and resolved compatibility issues for a stable build

Challenge · UI & Persistence Bugs

→ Used Streamlit session state and unique widget keys; added local persistent storage

Technologies Used

- Python
- Groq API + Llama 3
- LangChain
- ChromaDB + HuggingFace
- Streamlit + PyPDF

Business Impact

What these systems deliver in the real world

24 / 7

Availability

~80%

Time Saved on Doc Search

100%

Answers Source-Cited

- **Reduced support costs**

AI handles routine queries automatically — no extra staffing needed

- **Better customer experience**

Instant, accurate replies at any hour build trust and satisfaction

- **Faster decisions**

Leaders get document answers in seconds, not hours

- **Scales at no extra cost**

Handle far more volume without hiring more people

Thank You